# Efficient model customization with Microsoft Foundry

# GLOBAL AZURE PUGLIA 2026

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CODE architects — Tomorrow's software, today

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# What we will cover today

## Unlock business value with fine-tuning

Model customization is critical when you need to improve the quality, accuracy, and operation costs for your agentic AI apps

## Fine-tuning in Microsoft Foundry

When is fine-tuning the right choice for me?
Why is Microsoft Foundry the right platform to use?

## Fine-tuning for Agentic Applications

Why do models struggle with multi-turn tool calling?
How does Fine-Tuning improve agent performance?

**Demo: Basic Fine-Tuning**
Understand supervised fine-tuning

**Demo: Synthetic Data**
Get high-quality datasets with less effort

**Demo: Custom Grader**
Write evaluators to test target criteria

**Demo: Agentic Fine-Tuning**
SFT + Distillation to improve tool calling

## Summary

Model Customization – unlocks business value for enterprise
Microsoft Foundry - streamlines fine-tuning for AI developers

# Unlock business value with model optimization

# Why does Zava need model customization?

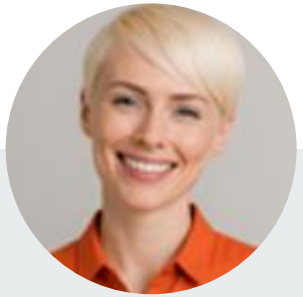
**Bruno Zhao**

Zava Customer

Help me find the right product for DIY project. Who can I talk to?

**Make it helpful**

Customize Cora's tone & response format to be polite, helpful & conversational



**Robin Counts**

Retail Store Manager

Help me build loyalty. Can we ensure its responses are valid?

**Make it accurate**

Improve Cora's use of tool selection, parameter propagation, policy adherence

**Kian Lambert**

App Dev Manager

Help me  save on operating costs. Can I teach a cheaper model this task?

**Make it cost-effective**

Distill Cora's task-specific behaviors into smaller models without losing accuracy

# The Zava AI Engineer's customization journey

I'm painting my living room wall. What paint should I buy

+

**Basic prompt engineering**

- Tone and style
- Example responses
- Intent mapping

**Retrieval/RAG**

- Query Extraction
- Personalization
- Inventory retrieval

+

**Fine-tune the LLM to:**

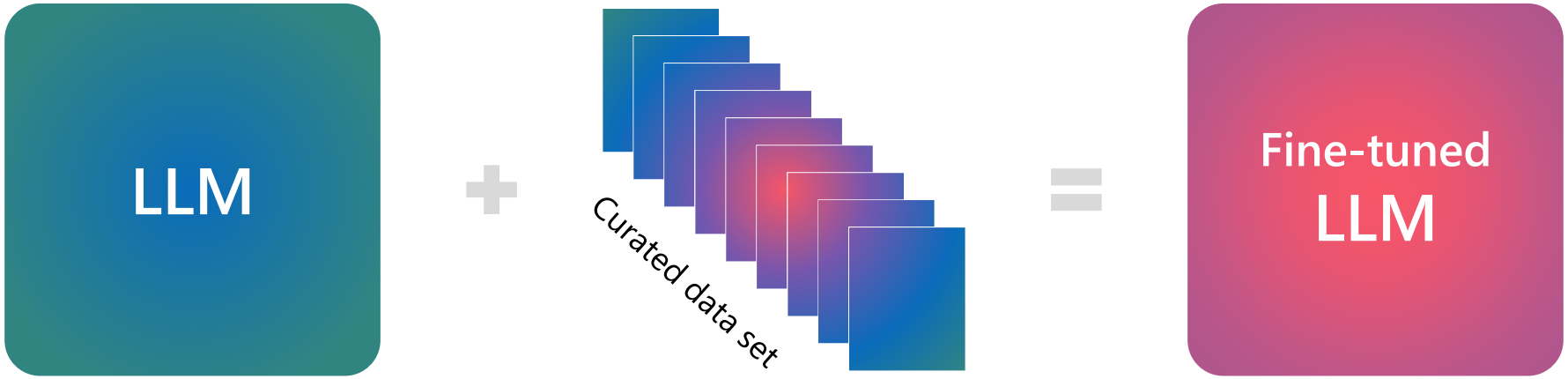
- Reduce the length of your prompt
- Show not tell the model how to behave
- Improve the accuracy when you look up information
- Improve the model's handling of retrieved data

=

I recommend our Eggshell Paint Would you like to know more about color choices

**User input** | **Context engineering** | **Model adaptation** | **Desired Output**

# What does fine-tuning mean?




Fine-tuning refers to **customizing a pre-trained LLM** with additional training **on a specific task or new dataset** for enhanced performance, new skills, or improved accuracy

# Why should Zava consider fine-tuning?

## Improve quality

- Domain-specific optimization
- Task-specific optimization

## Reduce cost

- Reduced token consumption
- Efficient resource utilization

## Reduce latency

- Smaller models, faster response
- Shorter prompts, improve response

**Example: Zava has a domain-specific focus (retail) and task-specific focus (question-answering). Let's think about how fine-tuning can help optimization**

# Customization options in Microsoft Foundry

# Fine-tuning in Foundry

## We're making every aspect of fine-tuning better

### No Data? No problem!

Synthesize training data from documents and code with synthetic data generation

Public Preview

### Train for Less!

50% discount with Dev Training tier – and higher quota

### Open Source Models

Ministral, OSS-20B, Llama 3.3 70B, and Qwen3 32B – in the same UX

Public Preview

### Agentic RFT

Fine-tune for tool use in GPT-5 chain of thought reasoning

Private Preview

AI.Azure.com

# What are my fine-tuning options in Microsoft Foundry?

**Supervised fine-tuning**

Module learns from examples
Ex: Content generation task

**Reinforcement fine-tuning**

Use grader to reinforce CoT
Ex: Reasoning tasks

**Model distillation**

Transfer learning to cheaper model
Optimize for COST

**Vision fine-tuning**

Improve image understanding
Ex: ClassificationTask

**Preference fine-tuning**

Provide good & bad examples
Ex: Tone adaptation

**Hybrid fine-tuning**

Improve model use of RAG context
Optimize for PRECISION

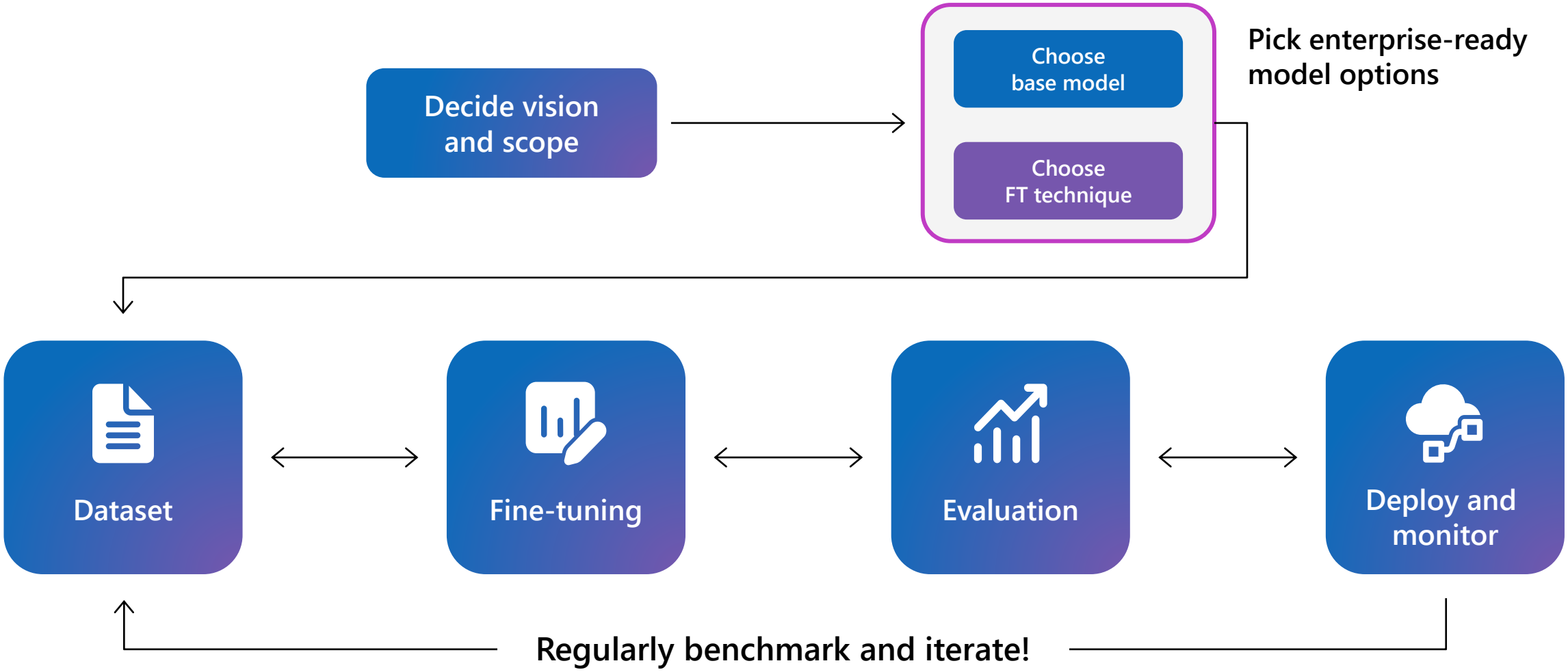
# Demo: Supervised fine-tuning in Zava

**Bruno Zhao**
Zava Customer

## Make it helpful

**Customize Cora's tone & response format to be polite, helpful & conversational**

# How can I fine-tune my model to customize the tone?



# Demo: Fine-tuning for Agentic Applications

## Make it accurate

**Improve Cora's use of tool selection, parameter propagation, policy adherence**

Robin Counts
Retail Store Manager

## Make it cost-effective

**Distill Cora's task-specific behaviors into smaller models without losing accuracy**

Kian Lambert
App Dev Manager

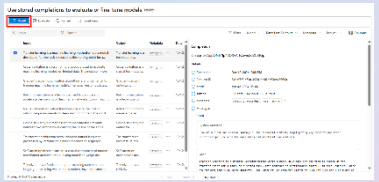
# Example: SFT+ Distillation for Tool Calling Accuracy in Microsoft Foundry

**Objective**

Teach a best-in-class model to call the right tools to solve complex business problems

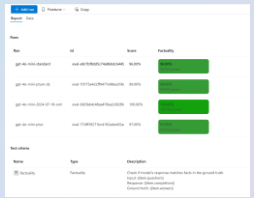
**Model: GPT 4.1-mini**

**Technique: Supervised fine-tuning**

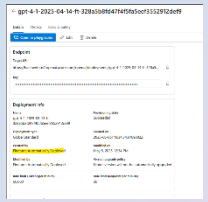


Data: Synthetic data generation

Train: Foundry UI & Python SDK



Evaluate: Foundry evals



Deploy: DevTier

Regularly benchmark and iterate!

Discovery Bank

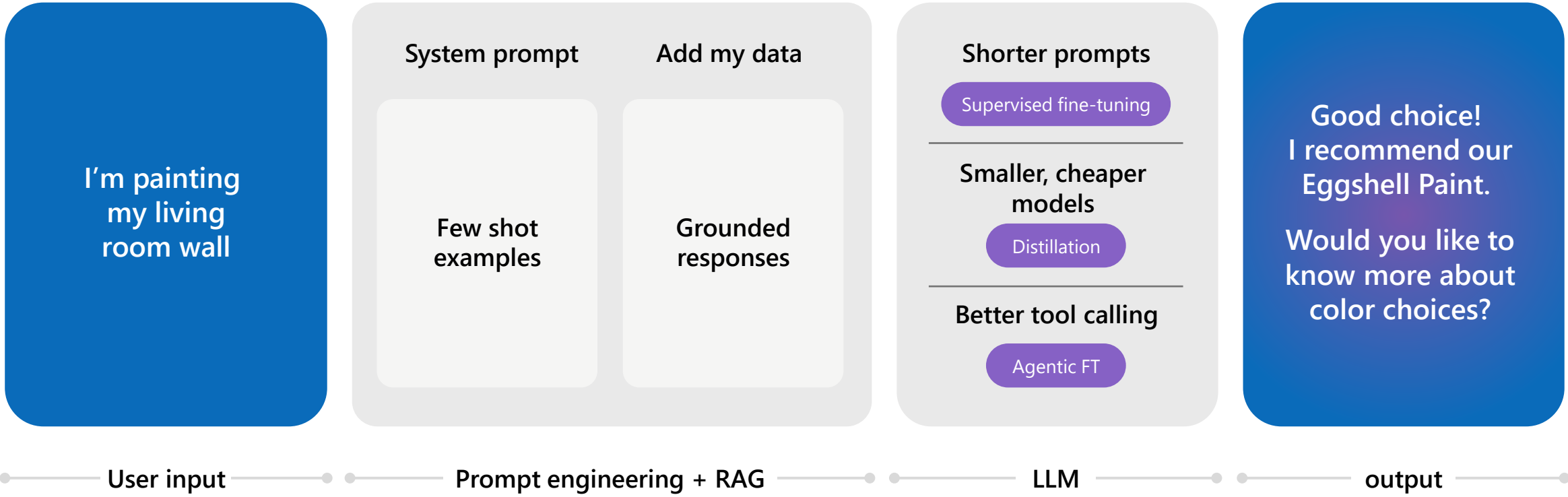
# Real world results from fine-tuning

*We consolidated three steps into one, response times that were previously five or six seconds came down to one and a half to two seconds on average. This approach made the system more efficient and the **50% reduction in latency** made conversations with Discovery AI feel seamless.*

—Stuart Emslie, Head of Actuarial and Data Science

# Summary & recap

# Recap: Model customization unlocks business goals



# Recap: Microsft Foundry makes fine-tuning seamless

### Model choice

The best models from the best providers

Choose serverless or managed compute

### Reliability

99.9% availability for Azure OpenAI models

Latency guarantees with PTU-M

### Foundry platform

Everything you need in one place: models, training, evaluation, deployments, and metrics

### Scalability

Start with low cost DevTier to experiment

Scale up with PTU-M for production workloads

## Join the Foundry Discord community

aka.ms/foundry/discord

## Visit session material

aka.ms/BRK443GHrepo

# Roma
# 15 Maggio 2026
# h 9.00

# Give us Feedback